**1. Differentiate between IPv4 and IPv6.**

| IPv4 | IPv6 |
| --- | --- |
| IPv4 has a 32-bit address length | IPv6 has a 128-bit address length |
| It Supports Manual and□DHCP address configuration | It supports Auto and renumbering address configuration |
| In IPv4 end to end, connection integrity is Unachievable | In IPv6 end-to-end, connection integrity is Achievable |
| It can generate 4.29·10□9□address space | The address space of IPv6 is quite large it can produce 3.4·10□38□address space |
| The Security feature is dependent on the application | IPSEC is an inbuilt security feature in the IPv6 protocol |
| Address representation of IPv4 is in decimal | Address representation of IPv6 is in hexadecimal |

| IPv4 | IPv6 |
|---|---|
| Fragmentation performed by Sender and forwarding routers | In IPv6 fragmentation is performed only by the sender |
| In IPv4 Packet flow identification is not available | In IPv6 packet flow identification are Available and uses the flow label field in the header |
| It has a broadcast Message Transmission Scheme | In IPv6 multicast and anycast message transmission scheme is available |
| In IPv4 Encryption and Authentication facility not provided | In IPv6□Encryption and Authentication are provided |
| IPv4□s □IP addresses are divided into five different classes. Class A , Class B, Class C, Class D , Class E. | IPv6 does not have any classes of the IP address. |
| IPv4 supports VLSM(□Variable Length subnet mask ). | IPv6 does not support VLSM. |

| IPv4 | IPv6 |
|---|---|
| **Example of IPv4:**<br>**□66.94.29.13** | **Example of IPv6:**<br>**2001:0000:3238:DFE1:0063:0000:0000:FEFB** |

**2. Compare TCP and UDP.**

| Feature | TCP | UDP |
|---|---|---|
| Connection-oriented | Yes, establishes a connection before data transfer. | No, does not establish a connection. |
| Reliability | Provides reliable data transmission (acknowledgments, retransmission). | No guarantees; unreliable data transmission. |
| Flow Control | Yes, uses mechanisms like sliding window for flow control. | No flow control. |
| Error Checking | Error detection and correction (checksums). | Error detection only (checksums). |

| Feature | TCP | UDP |
|---|---|---|
| Ordering | Ensures data is received in the same order it was sent. | No guarantee of order. |
| Speed | Slower due to error handling, retransmissions, etc. | Faster because there is no overhead for connection management. |
| Use Cases | File transfers, web browsing, email, etc. (where reliability is critical). | Streaming, online gaming, DNS, etc. |
| Header Size | 20 bytes | 8 bytes |
| Congestion Control | Yes, uses algorithms like TCP congestion control. | No congestion control. |

## 3. Draw and explain the IPv6 protocol format.

**IPv6 Protocol Format**

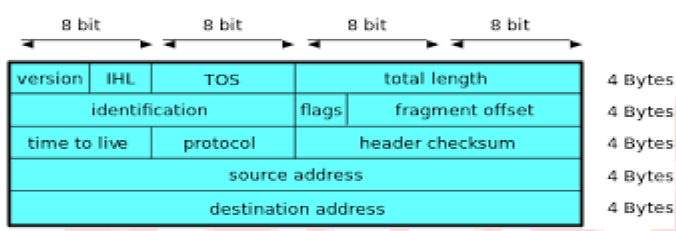
IPv6 (Internet Protocol version 6) is the most recent version of the Internet Protocol, designed to replace IPv4. It uses 128-bit addresses, which allows for a significantly larger number of devices to be addressed. Here's an explanation of the IPv6 packet format:

**IPv6 Header Format**

| Field Name | Length | Description |
|---|---|---|
| **Version** | 4 bits | Specifies the version of the IP protocol (for IPv6, it is always 6). |
| **Traffic Class** | 8 bits | Used for QoS (Quality of Service) to specify the priority of the packet. |
| **Flow Label** | 20 bits | Used for labeling packets belonging to the same flow (e.g., real-time data). |

| Field Name | Length | Description |
|---|---|---|
| **Payload Length** | 16 bits | Specifies the length of the payload (data) in bytes. |
| **Next Header** | 8 bits | Indicates the type of the next header (e.g., TCP, UDP, ICMP). |
| **Hop Limit** | 8 bits | Decrements by 1 as the packet traverses each router; when it reaches 0, the packet is discarded. |
| **Source Address** | 128 bits | The IPv6 address of the source node. |
| **Destination Address** | 128 bits | The IPv6 address of the destination node. |

The IPv6 header format is more streamlined compared to IPv4, as it removes the need for certain fields that were present in the IPv4 header, such as checksum and fragmentation.




- **Version** (4 bits): This field is set to 6 for IPv6 packets.
- **Traffic Class** (8 bits): It's used to define the quality of service for a packet.
- **Flow Label** (20 bits): It helps manage packets from the same flow (e.g., VoIP or video conferencing).
- **Payload Length** (16 bits): Defines the size of the payload that follows the IPv6 header.
- **Next Header** (8 bits): Identifies the protocol used in the data portion (such as TCP, UDP, etc.).
- **Hop Limit** (8 bits): Sets the maximum number of hops (routers) the packet can pass through before being discarded.
- **Source Address** (128 bits) and **Destination Address** (128 bits): Represent the sender's and receiver's IPv6 addresses.
- **Data/Payload**: The actual data being transmitted. Its length is indicated by the Payload Length field.

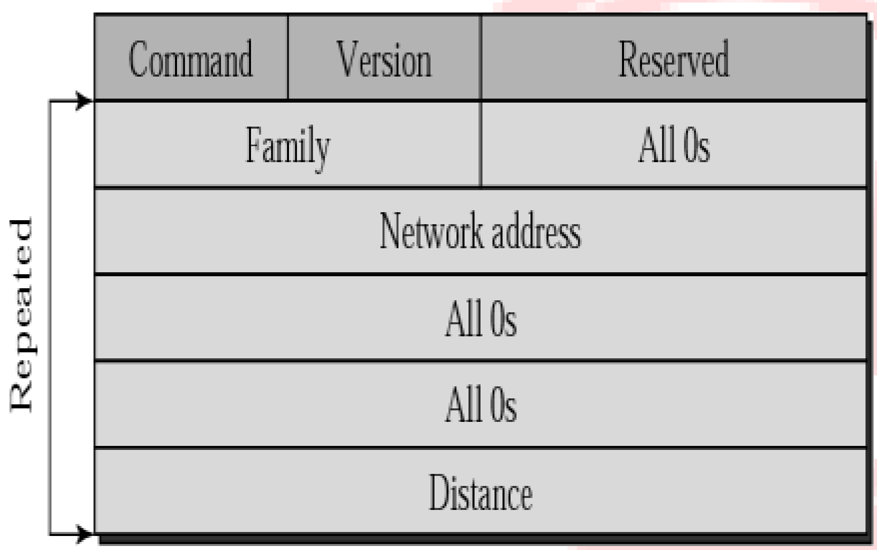
## 4. State the use of six flags in the TCP header.

There are 6, 1-bit control bits that control connection establishment, termination, abortion, flow control etc..
1. URG: The urgent pointer is valid if it is 1.
2. ACK: ACK bit is set to 1 to indicate the acknowledgement member is valid.
3. PSH: The receiver should pass this data to application as soon as possible.
4. RST: This flag is used to reset connection.

5. SYN: Synchronize sequence number to initiate a connection.
6. FIN: It is used to release connection

## 5. Describe the RIP message format.

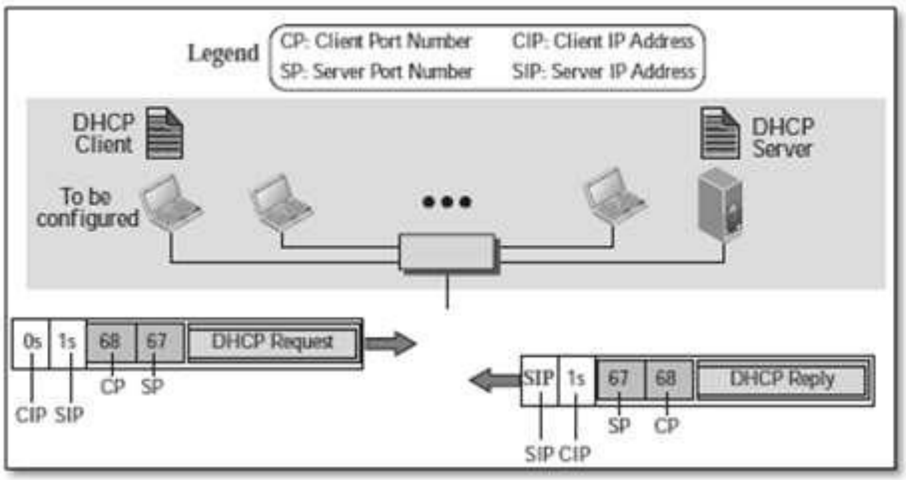
•RIP is routing protocol based on Distance Vector Routing algorithm which is an intra domain (interior) routing protocol used inside an autonomous system.
•The metric used by RIP is the distance which is defin
ed as the number of links (networks)
that have to be used to reach the destination. For this reason, the metric in RIP is called a hop count.
•Infinity is defined as 16, which means that any route in an autonomous system using RIP cannot have more than 15
hops.
The next node column defines the address of the router to which the packet is to be sent to reach its destination.




**Command: 8-bit** The type of message: request (1) or response (2)
• **Version: 8-bit** Define the RIP version
• **All 0s**
This field is not actually used by RFC 1058 RIP; it was added solely to provide backward compatibility with pre-standard varieties of RIP. Its name comes from its defaulted value, zero.
• **Family:** 16-bit field defines the family of the protocol used. For TCP/IP, value is 2
• **IP Address Network Address:**
14 bytes n Defines the address of the destination network and
14 bytes for this field to be applicable to any protocol. However, IP currently uses only 4 bytes, the rest are all 0s
**Distance:** 32 -bit field defines the hop count from the advertising router to the destination network

## 5. Describe DHCP with its operation, including static and dynamic allocation.

DHCP is based on a client-server model and based on discovery, offer, request, and ACK.DHCP client and server can either be on the same network or on different networks. DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so it can communicate using IP. DHCP automates and centrally manages these configurations rather than requiring network administrators to manually assign IP addresses to all network devices. DHCP can be implemented on small local networks, as well as large enterprise networks. DHCP assigns new IP addresses in each location when devices are moved from place to place, which means network administrators do not have to manually configure
each device with a valid IP address or reconfigure the device with a new IP address if it moves to a new location on the network.




In this case, the operation can be described as follows:
1. The DHCP server issues a passive open command on UDP port number 67 and waits for a client.
2. A booted client issues an active open command on port number 68. The message is encapsulated in a UDP user datagram, using the destination port number 67 and the source port number 68.
3. The server responds with either a broadcast or a unicast message using UDP source port number 67 and destination port number 68.

**STATIC ALLOCATION**

The static allocation method is very popular in modern ISP networks, which do not use dial-up methods. With the static allocation, the DHCP sever keeps a database with all clients' LAN MAC addresses and gives them an IP address only if their MAC address is in the database. This way, the clients can be sure that they will be getting the same IP address every time.

**DYNAMIC ALLOCATION**

When the DHCP server is configured to use dynamic allocation, this means that it uses a lease policy. This way, when an assigned IP address from the available pool is no longer used, it will be transferred back to the pool, making it available for someone else to use. The advantage of this method is that the IP addresses are used to their maximum - as soon as they are no longer used by the client, they are instantly made available to others. The disadvantage of this method is that a client will always have a random IP address.

## 6. Explain the difference between distance vector and link state routing

| Sr. No. | Distance Vector Routing | Link State Routing |
|---|---|---|
| 1 | Routing tables are updated by exchanging information with the neighbours. | Complete topology is distributed to every router to update a routing table. |
| 2 | It update full routing table. | It updates only link states. |
| 3 | It uses Bellman-Ford algorithm | It uses Dijkstra algorithm. |
| 4 | Distance Vector routing doesn't have any hierarchical structure. | Link state routing works best for hierarchical routing design. |
| 5 | CPU and memory utilization is lower than Link state routing. | Higher utilization of CPU and memory than distance vector routing. |
| 6 | Bandwidth required is less due to local sharing, small packets and no flooding. | Bandwidth required is more due to flooding and sending of large link state packets. |
| 7 | Example protocols are RIP and IGRP. | Example protocols are OSPF and IS-IS. |
| 8 | Slow convergence. | Fast convergence. |
| 9 | Summarization is automatic | Summarization is manual. |
| 10 | Easier to configure | Harder to configure |
| 11 | Count to infinity problem | No count to infinity problem |

## 7. Describe the architecture of an email system using four scenarios.

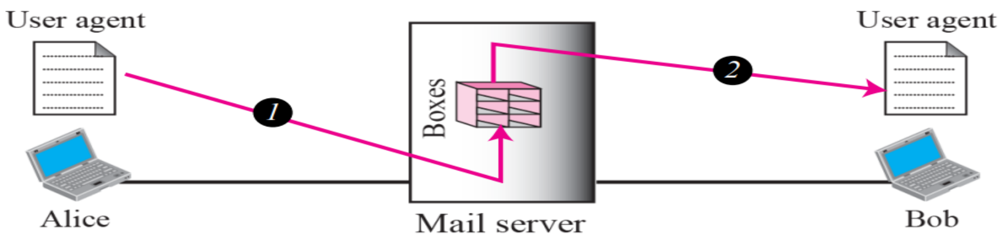
To explain the architecture of e-mail, we give four scenarios. We begin with the simplest situation and add complexity as we proceed. The fourth scenario is the most common in the exchange of e-mail.

**TCP/IP Protocol Suite 2 Topics Discussed in the Section**

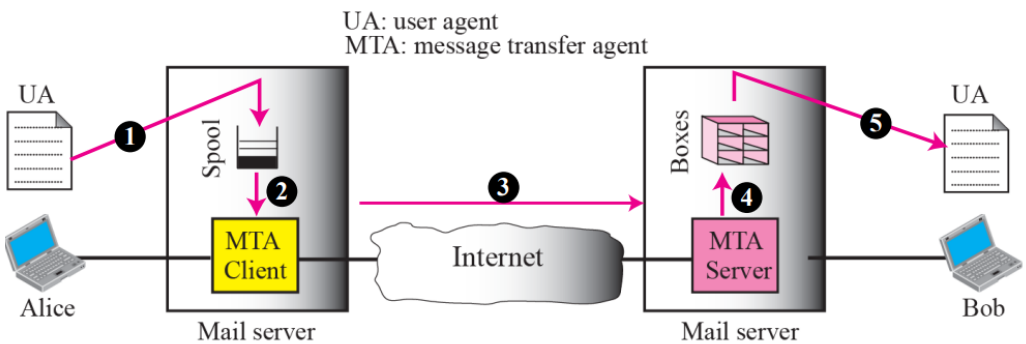
• First Scenario
• Second Scenario
• Third Scenario
• Fourth Scenario

• First Scenario

When the sender and the receiver of an e-mail are on the same mail server, we need only two user agents.

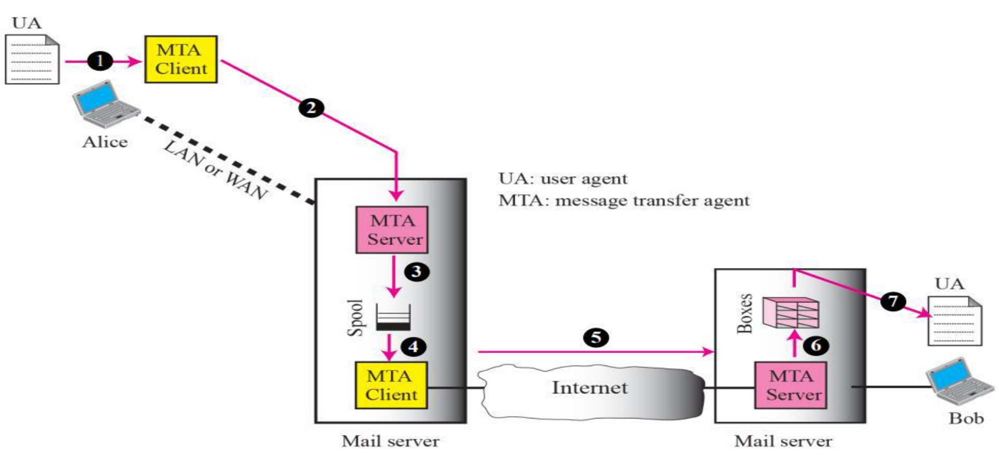



Second Scenario
When the sender and the receiver of an e-mail are on different mail servers, we need two UAs and a pair of MTAs (client and server).



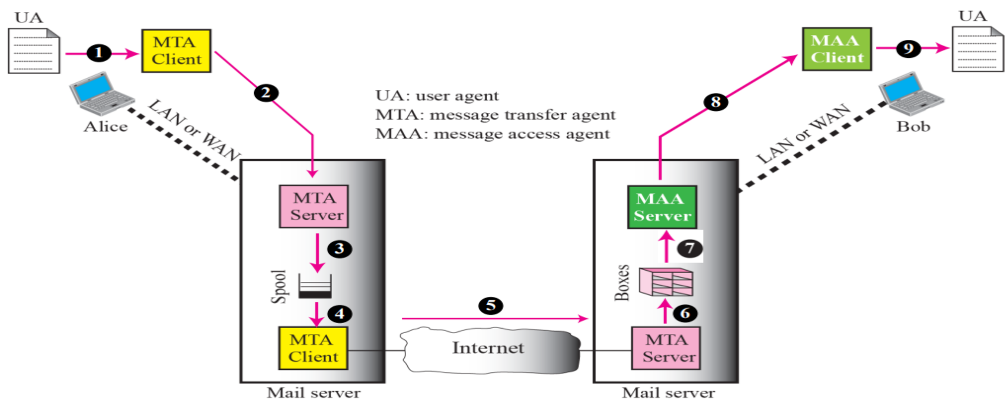

Third Scenario

When the sender is connected to the mail server via a LAN or a WAN, we need two UAs and two pairs of MTAs (client and server).



Fourth Scenario

When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server). This is the most common situation today.




## 9. Differentiate between RIP and OSPF routing protocols.

## Ans:-

| RIP | OSPF |
|---|---|
| RIP Stands for Routing Information Protocol. | OSPF stands for Open Shortest Path First. |
| RIP works on the Bellman-Ford algorithm. | OSPF works on Dijkstra algorithm. |
| It is a Distance Vector protocol and it uses the distance or hops count to determine the transmission path. | It is a link-state protocol and it analyzes different sources like the speed, cost and path congestion while identifying the shortest path. |
| It is used for smaller size organizations. | It is used for larger size organizations in the network. |
| It allows a maximum of 15 hops. | There is no such restriction on the hop count. |
| It is not a more intelligent dynamic routing protocol. | It is a more intelligent routing protocol than RIP. |
| The networks are classified as areas and tables here. | The networks are classified as areas, sub-areas, autonomous systems, and backbone areas here. |
| Its administrative distance is 120. | Its administrative distance is 110. |
| RIP uses UDP(User Datagram Protocol) Protocol. | OSPF works for IP(Internet Protocol) Protocol. |
| It calculates the metric in terms of Hop Count. | It calculates the metric in terms of bandwidth. |
| In RIP, the whole routing table is to be broadcasted to the neighbors every 30 seconds by the routers. | In OSPF, parts of the routing table are only sent when a change has been made to it. |
| RIP utilizes less memory compared to OSPF but is CPU intensive like OSPF. | OSPF device resource requirements are CPU intensive and memory |
| It consumes more bandwidth because of greater network resource requirements in sending the whole routing table. | It consumes less bandwidth as only part of the routing table is to send. |

**10. Explain the Bellman-Ford algorithm with a suitable example.**

**Ans:-** i. Bellman ford algorithm is a single-source shortest path algorithm.

ii. This algorithm is used to find the shortest distance from the single vertex to all the other vertices of a weighted graph.

iii. Various other algorithms are used to find the shortest path, like the Dijkstra algorithm.

iv. If the weighted graph contains the negative weight values, then the Dijkstra algorithm does not confirm whether it produces the correct answer or not.

v. Rule for the algorithm:

We will go on relaxing all the edges (n - 1) times where

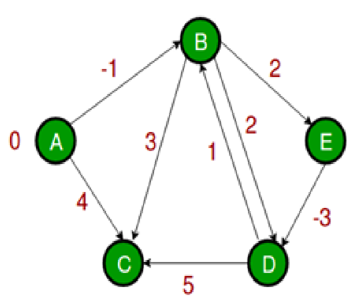
n = number of vertices

vi. Relaxing means:

If (d(u) + c(u , v) < d(v))

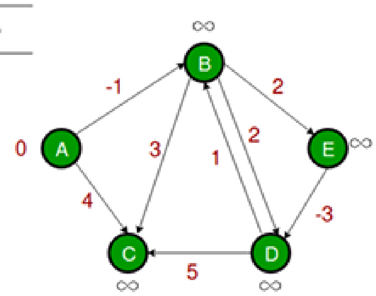
d(v) = d(u) + c(u , v)

vii. Consider the following example:-

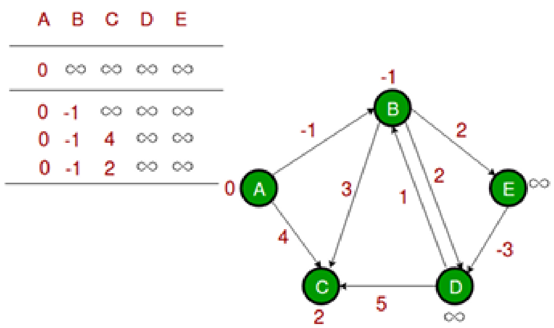



Step 1: Let the given source vertex be 0. Initialize all distances as infinite, except the distance to the source itself. The total number of vertices in the graph is 5, so all edges must be processed 4 times.

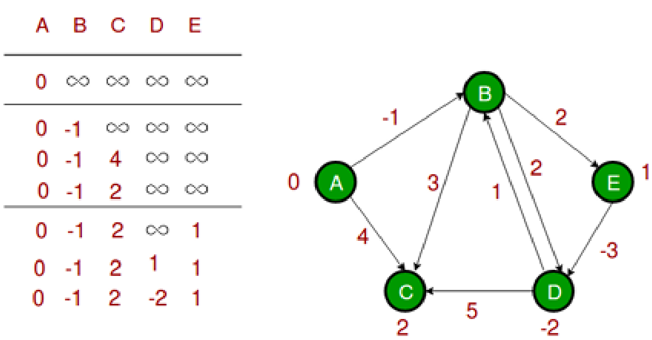
| A | B | C | D | E |
|---|---|---|---|---|
| 0 | ∞ | ∞ | ∞ | ∞ |



Step 2: Let all edges are processed in the following order: (B, E), (D, B),(B, D), (A, B), (A, C), (D, C), (B, C), (E, D). We get the following distances when all edges are processed the first time. The first row shows initial distances. The second row shows distances when edges (B, E), (D, B), (B, D) and (A, B) are processed. The third row shows distances when (A, C) is processed. The fourth row shows when (D, C), (B, C) and (E, D) are processed.



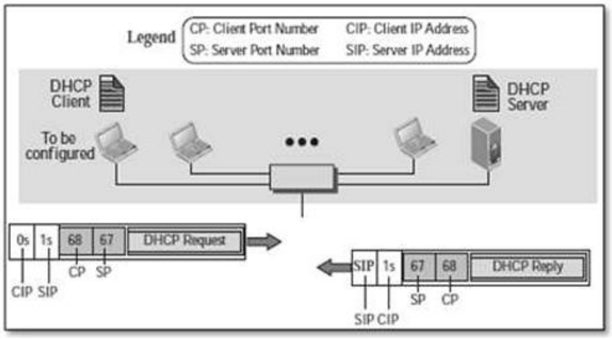

Step 3: The first iteration guarantees to give all shortest paths which are at most 1 edge long. We get the following distances when all edges are processed second time (The last row shows final values).




Step 4: The second iteration guarantees to give all shortest paths which are at most 2 edges long. The algorithm processes all edges 2 more times. The distances are minimized after the second iteration, so third and fourth iterations do not update the distances.

## 11. Describe DHCP operations when the DHCP client and server are on the same network.

**Ans:-** DHCP is based on a client-server model and based on discovery, offer, request, and ACK.DHCP client and server can either be on the same network or on different networks. DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so it can communicate using IP. DHCP automates and centrally manages these configurations rather than requiring network administrators to manually assign IP addresses to all network devices. DHCP can be implemented on small local networks, as well as large enterprise networks. DHCP assigns new IP addresses in each location when devices are moved from place to place, which means network administrators do not have to manually configure each device with a valid IP address or reconfigure the device with a new IP address if it moves to a new location on the network.




In this case, the operation can be described as follows:

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3. The server responds with either a broadcast or a unicast message using UDP source port number 67 and destination port number 68.

## 12. Explain the following address types of IPv6:

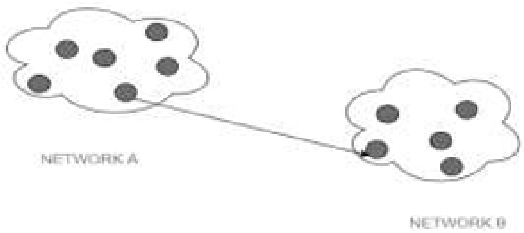
## - Unicast address

## - Multicast address

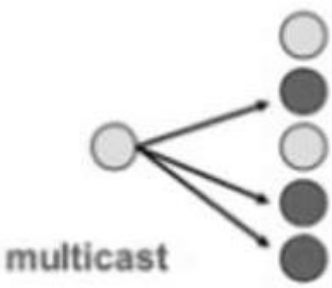
## - Anycast address

**Ans:-** The three types of IPv6 addresses are: unicast, anycast, and multicast.

a) **Unicast address**: This type of information transfer is useful when there is a participation of single sender and single recipient. So, in short you can term it as a one-to-one transmission. For example, a device having IP address 10.1.2.0 in a network wants to send the traffic stream (data packets) to the device with IP address 20.12.4.2 in the other network, then unicast comes into picture. This is the most common form of data transfer over the networks.




b) **Multicast address**: In multicasting, one/more senders and one/more recipients participate in data transfer traffic. In this method traffic recline between the boundaries of unicast (one-to-one) and broadcast (one-to-all). Multicast lets server's direct single copies of data streams that are then simulated and routed to hosts that request it. IP multicast requires support of some other protocols like IGMP (Internet Group Management Protocol), Multicast routing for its working. Also, in Classful IP addressing Class D is reserved for multicast groups.




c) Anycast address: An IPv6 anycast address is an address that is assigned to more than one interface (typically belonging to different nodes), where a packet sent to an anycast address is routed to the nearest interface having that address, according to the routing protocol's measure of distance. Anycast addresses, when used as part of a route sequence, permit a node to select which of several Internet service providers it wants to carry its traffic. This capability is sometimes called source selected policies. You implement this by configuring anycast addresses to identify the set of routers belonging to Internet service providers (for example, one anycast address per Internet service provider). You can use these anycast addresses as intermediate addresses in an IPv6 routing header, to cause a packet to be delivered by a particular provider or sequence of providers. You can also use anycast addresses to identify the set of routers attached to a particular subnet or the set of routers providing entry into a particular routing domain. You can locate anycast addresses from the unicast address space by using any of the defined unicast address formats. Thus, anycast addresses are syntactically indistinguishable from unicast addresses. When you assign a unicast address to more than one interface, that is, turning it into an anycast address, you must explicitly configure the nodes to which the address is assigned in order to know that it is an anycast address.

## 13. Construct a suitable diagram for each FTP command below to show its use:

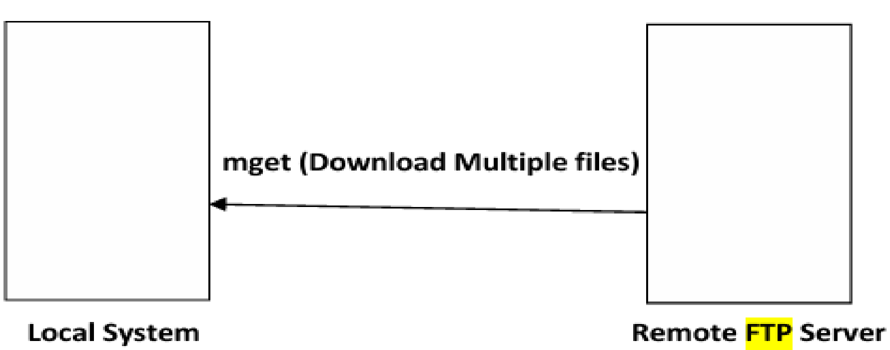
- `get`
- `mget`
- `put`
- `mput`

**Ans:-**

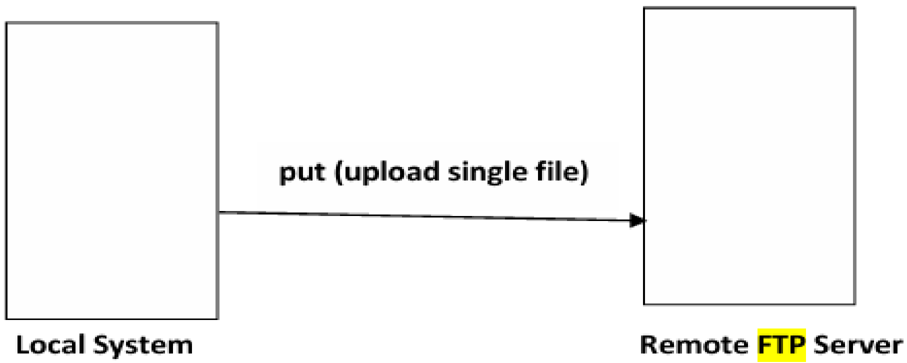
| Command | Purpose | Syntax |
|---|---|---|
| get | Download a single file | get<filename> |
| mget | Download multiple files | mget<filename1 filename2 filename3> |
| put | Upload a single file | put<filename> |
| mput | Upload multiple files | mput< filename1 filename2 filename3 > |

**i) get**

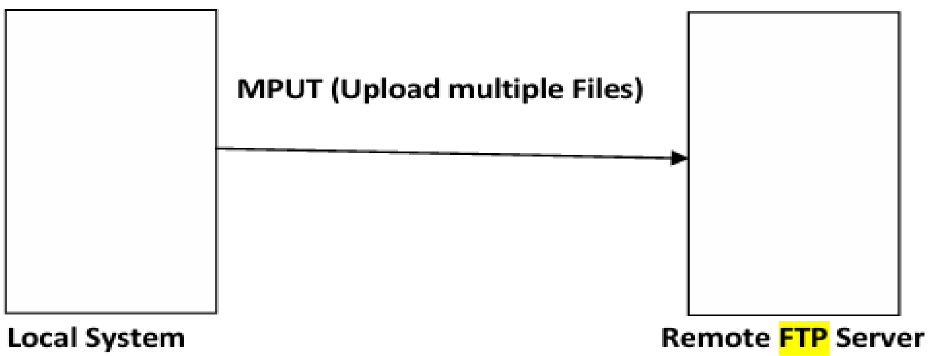

**ii) mget**



ii) put




iv)mput




## 14. Explain the working of the World Wide Web (WWW).

**Ans:-** The Web is a repository of information in which the documents, called web pages, are distributed all over the world and related documents are linked together.

The WWW today is a distributed client-server service, in which a client using a browser can access a service using a server. The service provided is distributed over many locations called sites. Each site holds one or more web pages. Each web page can contain some links to other web pages in the same or other sites.

• Simple web page has no links to other web pages.

• Composite web page has one or more links to other web pages. Each web page is a file with a name and address.

The web page is stored at the web server.

Each time a request arrives, the corresponding document is sent to the client.

## 15. List all four routing algorithms (briefly, only with definitions).

**Ans:-**

## 16. Elaborate on the need for the Domain Name System (DNS).

**Ans:-** 1. DNS ensures the internet is not only user-friendly but also works smoothly, loading whatever content we ask for quickly and efficiently.

2. It allows the user to access remote system by entering human readable device hostnames instead of IP address. It translates domain name into IP addresses so browser can load internet resources.

3. It translates human readable domain names into the numerical identifiers associated with networking equipment, enabling devices to be located and connected worldwide. Analogous to a network phone book, DNS is how a browser can translate a domain name (e.g., facebook.com) to the actual IP address of the server, which stores the information requested by the browser.

## 17. From the following, explain any two transition methods from IPv4 to IPv6:

### - Dual stack

### - Tunneling

### - Header translation

**Ans:-**

1. Dual Stack In this kind of strategy, a station has a dual stack of protocols run IPv4 and IPv6 simultaneously. To determine which version to use when sending a packet to a destination, the source host queries the DNS. If the DNS returns an IPv4 address, the source host sends an IPv4 packet. If the DNS returns an IPv6 address, the source host sends an IPv6 packet.

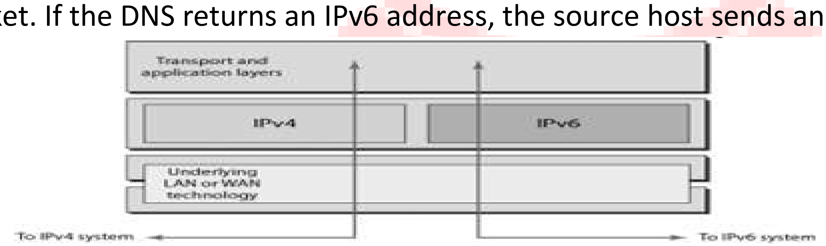



**Fig. Dual Stack**

2. Tunneling Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4. To pass through this region, the packet must have an IPv4 address. So, the IPv6 packet is encapsulated in an IPv4 packet when it enters the region. To make it clear that the IPv4 packet is carrying an IPv6 packet as data the protocol value is set to 41.

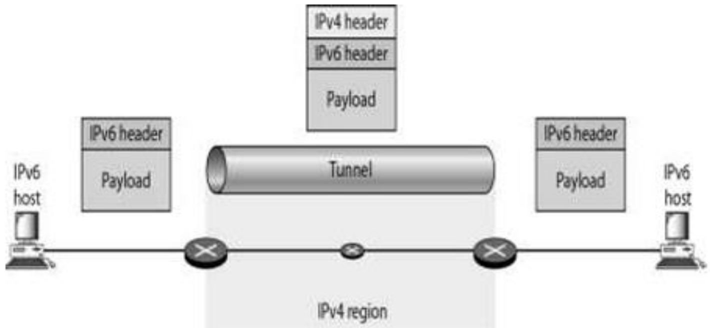


**Fig Tunneling**

3. Header Translation In this case, the header format must be totally changed through header translation. The header of the IPv6 packet is converted to an Ipv4 header see figure.

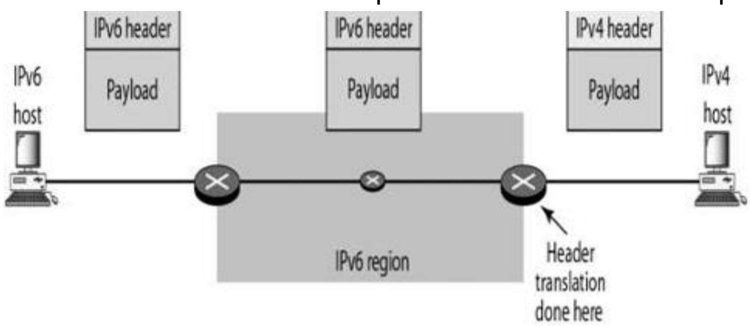


**Fig. Header Translation**

## 18. Explain the working of TELNET.

**Ans:- TELNET: TELNET is an abbreviation for TErminaLNETwork. It is the standard TCP/IP protocol for virtual terminal service.**

## TELNET Working:

• TELNET is a client-server application that allows a user to log on to a remote machine, giving the user access to the remote system.

• The user sends the keystrokes to the terminal driver, where the local operating system accepts the characters but does not interpret them.

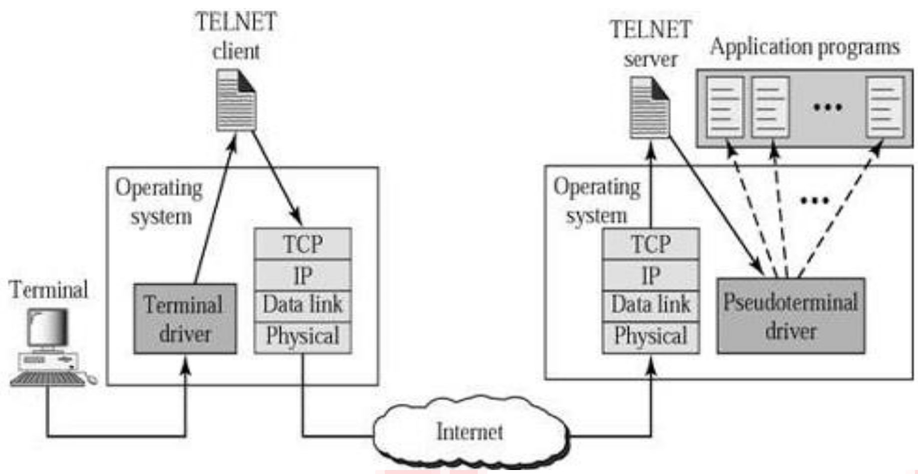
• A terminal driver correctly interprets the keystrokes on the local terminal or terminal emulator. The characters are sent to the TELNET client, which transforms the characters to a universal character set called network virtual terminal (NVT) characters and delivers them to the local TCP/IP protocol stack.

• The commands or text, in NVT form, travel through the Internet and arrive at the TCP/IP stack at the remote machine.

• Here the characters are delivered to the operating system and passed to the TELNET server, which changes the characters to the corresponding characters understandable by the remote computer.

• However, the characters cannot be passed directly to the operating system because the remote operating system is not designed to receive characters from a TELNET server: It is designed to receive characters from a terminal driver.

• A piece of software called a pseudo terminal driver is added which pretends that the characters are coming from a terminal.




## 19. Distinguish between SMTP and POP3 protocols (four points each).

**Ans:-**

| Basis of comparison | Static Routing | Dynamic Routing |
|---|---|---|
| Configuration | Manually done | Automatically done |
| Routers | Routing location by hand typed | Dynamically fill all locations |
| Routing algorithms | Does not support complex algorithm | Supports more complex algorithm for routing purposes |
| Used in | In small networks | In large networks |
| Filure of links | Link failure disturb rerouting | Link failure doesnot disturb the rerouting |
| Security | More secure because no advertisement send with data | Less secure because sending multicast and broadcasts |
| Routing Protocol | No routing protocols are added in the routing process | Routing protocols such as RIP EIGRP etc are included in all routing process |
| Extra resources | There is no extra resource like memory and CPU. | It requires resource like memory and CPU etc. |

**20. List any two services offered by UDP.**

**Ans:-**

**21. Given the following UDP header in hexadecimal format: `BC82D00D002B001D`, obtain:**

- **Source port number**
- **Destination port number**
- **Total length**
- **Packet direction**

**Ans:-**

Considering hexadecimal format as:

**BC82D00D002B001D**

The UDP header has four parts, each of two bytes.

That means we get the following interpretation of the header.

i) Source port number = $BC82_{16}$ = 48258

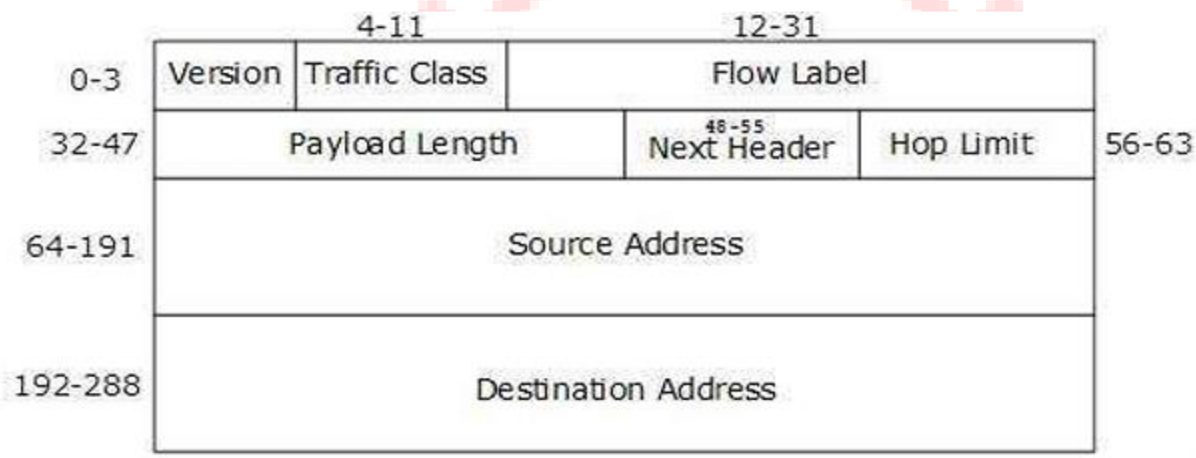
ii) Destination port number = $D00D_{16}$= 53261

iii) Total length = $002B_{16}$ = 43 bytes

iv) Packet direction: The provided dump does not contain information about the packet direction. The UDP header alone does not specify the direction.

## 22. Describe the packet format of IPv6.

**Ans:-**




**Version (4-bits):** It represents the version of Internet Protocol, i.e. 0110. Traffic Class (8-bits): These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet.

**Flow Label (20-bits):** This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

**Payload Length (16-bits):** This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data.

**Next Header (8-bits):** This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

**Hop Limit (8-bits):** This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link. When the field reaches 0 the packet is discarded.

**Source Address (128-bits):** This field indicates the address of originator of the packet.

**Destination Address (128-bits):** This field provides the address of intended recipient of the packet.

**Extension Headers** :When Extension Headers are used, IPv6 Fixed Header's Next Header field points to the first Extension Header. If there is one more Extension Header, then the first Extension Header's 'Next-Header' field points to the second one, and so on. The last Extension Header's 'Next-Header' field points to the Upper Layer Header. Thus, all the headers points to the next one in a linked list manner.

## 23. Describe the architecture of an email system over no secure channel..

## 24. Explain the process of resolving a hostname into an IP address using DNS.

### Ans:- Address using DNS.

You can find the hostname of any computer with a public IP address by passing the address to any Domain Name System (DNS) server. However, since the computers on a small business network have private IP addresses, you can only discover their hostnames if the network has a local DNS server. To discover the hostname of a computer with a private IP address and no local DNS server, you need to use a Windows utility to query the host itself.

## Querying DNS

1. Click the Windows Start button, then "All Programs" and "Accessories." Right-click on "Command Prompt" and choose "Run as Administrator."

2. Type "nslookup %ipaddress%" in the black box that appears on the screen, substituting %ipaddress% with the IP address for which you want to find the hostname.

3. Find the line labeled "Name" underneath the line with the IP address you entered and record the value next to "Name" as the hostname of the computer

## 25. Explain how TCP connections are established using the three-way handshake. What happens when two hosts simultaneously try to establish a connection?

**Ans:-** TCP uses a Three way handshaking mechanism to establish a connection between client and server machines.

The three steps in three way handshaking mechanism are as follows.

### SYN:

The client sends the first segment, a SYN segment, in which only the SYN flag is set. This segment is for synchronization of sequence numbers.

### SYN + ACK

The server sends the second segment, a SYN +ACK segment, with 2 flag bits set.

### ACK

The client sends the third segment. This is just an ACK segment. It guarantees the completion of three way handshaking.